ETSI/TC SMG
Released by : ETSI/PT 12
Release date: February 1992

RELEASE NOTE

Recommendation GSM 02.02

Bearer Services supported by a GSM PLMN

Previously distributed version : 3.2.0 (Updated Release 1/90)
New Released version February 92 : 3.2.0 (Release 92, Phase 1)

1. Reason for changes

No changes since the previously distributed version.

ETSI-GSM
Technical
Specification

GSM 02.02

Version 3.2.0

UDC: 621.396.21

Key words: European Digital Cellular Telecommunications System, Global System for Mobile Communications (GSM)

European digital cellular telecommunication system (phase 1);

Bearer Services supported by a GSM PLMN

ETSI

European Telecommunications Standards Institute

ETSI Secretariat: B.P.152 . F - 06561 Valbonne Cedex . France

TP. + 33 92 94 42 00 TF. + 33 93 65 47 16 Tx. 47 00 40 F

Copyright European Telecommunications Standards Institute 1992.
All rights reserved.

No part may be reproduced or used except as authorised by contract or other written permission. The copyright and the foregoing restriction on reproduction and use extend to all media in which the information may be embodied.

ETSI GSM

Released by: ETSI PT12

Date: February 1992

Recommendation: GSM 02.02

Title: BEARER SERVICES SUPPORTED BY A GSM PLMN

List of Contents:

0. Scope

1. Framework for describing bearer services

2. List of bearer service attributes

3. List of individual bearer services

4. Description of individual bearer services

5. List of bearer service categories

6. List of bearer services

Original language: English

Number of pages: 38

0. SCOPE

The purpose of this recommendation is to describe and define a recommended set of bearer services, to be provided to GSM PLMN subscribers by a GSM PLMN, in connection with other networks. This Recommendation should also be used as a basis for defining the corresponding required mobile network capabilities which are specified by means of the "GSM PLMN connection type" concept, defined in GSM Recommendation 03.10.

Bearer services not included in this recommendation should not be introduced unilaterally by a mobile network operator, if the provision of this service requires that the GSM signalling Recommendations are modified.

The definitions of the quality of service attributes are given in recommendation 02.08.

1. FRAMEWORK FOR DESCRIBING BEARER SERVICES

Bearer services are described by attributes, which are intended to be independent. These attributes are described and defined in recommendation GSM 02.01. They are grouped into four categories :

i) Information transfer attributes, which characterise the network capabilities for transferring information from one user access point in a GSM PLMN (originating PLMN), as defined in recommendation GSM 02.01 and GSM 04.02 to an appropriate [user] access point of a terminating network as defined in GSM 02.01 (the terminating network may include another GSM PLMN or the originating GSM PLMN).

ii) Access attributes, which describe the means for accessing network functions or facilities as seen at the access point in the PLMN (see GSM 02.01).

iii) Interworking attributes, which describe properties of the terminating network and its access point. The terminating network may include another GSM PLMN or the originating PLMN (see GSM 02.01).

iv) General attributes, which deal with the service in general.

Figure 1/GSM 02.02 shows the relation between the groups of attributes and their fields of applicability.

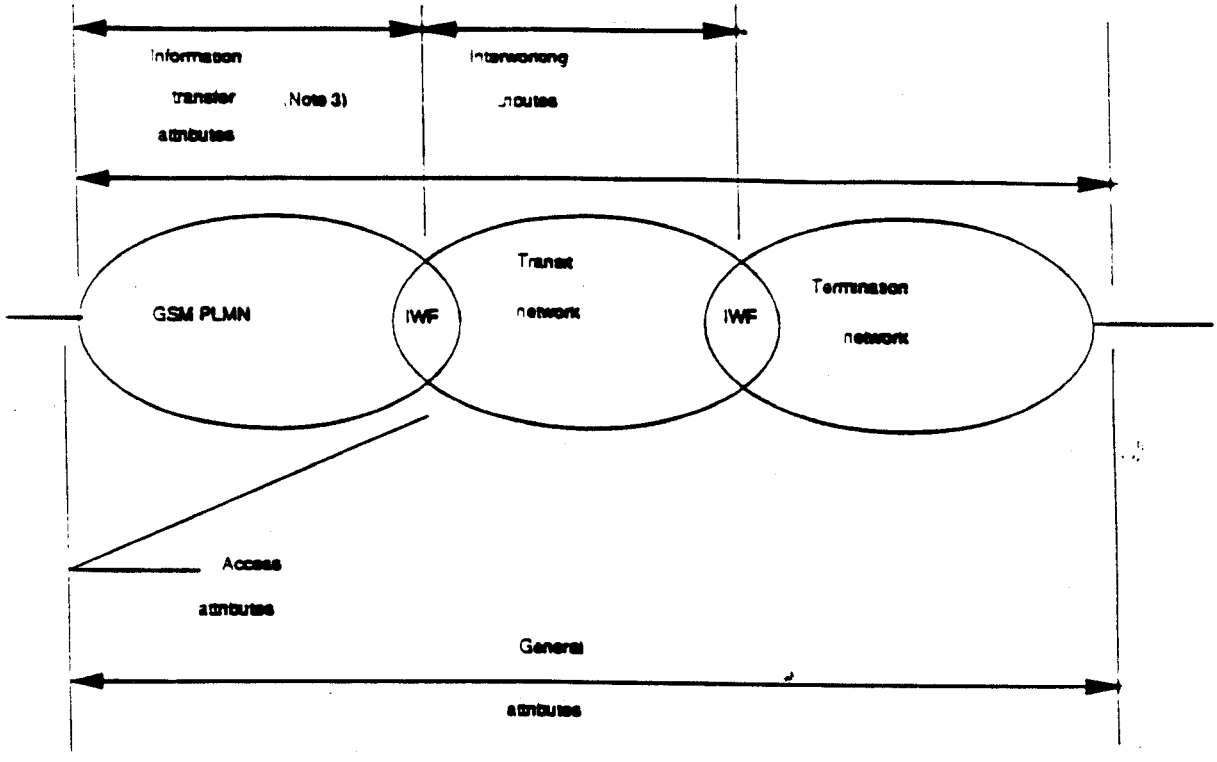

Fig. 1/GSM 02.02. Relation between the groups of attributes and fields of applicability

Note 1: A transit network may not exist for a bearer service.

Note 2: Communication may be established from both ends in principle.

Note 3: The information transfer and access attributes associated with a bearer service end to end communication relates to a direct peer-to-peer communication of TE to TE, or TE to a network gateway (supporting for example PSTN interworking).

2. LIST OF BEARER SERVICE ATTRIBUTES

Table 1/GSM 02.02 gives the list of the attributes and the principles for categorizing bearer services. For the definitions and possible values of these attributes, see recommendation GSM 02.01.

		Dominant attributes	Secondary attributes	Qualifying attributes
		Define bearer service category	Define bearer service within a category	Further qualify a bearer service
1 Information transfer	1 Information transfer mode	X		
	2 Information transfer rate	see note 1		
	3 Information transfer capability	X		
	4 Structure	X		
	5 Establishment of communication			X
	6 Communication configuration			X
	7 Symmetry			X
2 Access attributes	8 Access channel and rate			X
	9.1 Signalling access			X
	9.2 Information access		X	
	9.3 Information access structure		X	
3 Interworking with terminating network	12 Interworking possibilities		X	
4 General	10 Supplementary services			X
	11 Quality of service			X
	13 Operational commercial			X

Table 1/GSM 02.02. List of attributes for bearer services

Note: Within the GSM PLMN the "Information transfer rate" attribute is not indicated, this is because the user may access the PLMN at either an "S" or "R" reference point. In addition, the "Information transfer rate" at other reference points within the PLMN assumed or other wise may be different.

The description of the bearer services is based upon the list of attributes as given in table 1/GSM 02.02. Attributes nos. 1,2,3 and 4 are called "Dominant attributes". They are used to identify a particular bearer service category.

Attributes nos. 9.2, 9.3 and 12 are called "Secondary attributes". They are used to identify particular bearer services within an individual bearer service category.

All other attributes are used to further specify and qualify a bearer service within an individual bearer service category. They are called "Qualifying attributes".

3. LIST OF BEARER SERVICE CATEGORIES AND BEARER SERVICES

Table 2/GSM 02.02 gives the list of bearer service categories and the associated dominant attributes.

Table 3/GSM 02.02 Presents an overview of the bearer services with their associated dominant and secondary attributes. It is the addition of the secondary attributes which differentiate the bearer service categories from the bearer services. These secondary attributes primarily relate to the users interface at the mobile station e.g 9.6 kbit/s V32 synchronous and the interworking function required to the differing networks e.g PSTN, ISDN, PLMN, etc.

The "bearer service name" column is the generic name for the bearer services with the same dominant and secondary attributes. The "bearer service category" number column, indicates the number of the bearer service category the bearer service belongs to. All bearer services associated with a bearer service category are described in Annex A, in the data sheet numbered by the bearer service category number in Table 2.

The "Quality of service attribute" column refers to two different values of quality of service (Transparent T and Non-Transparent NT) as described in recommendation 02.08.

Table 4/GSM 02.02 lists the access interfaces at the mobile station and where interworking is required, the access to the terminating network.

Table 5/GSM 02.02 lists the bearer services with specific terminating networks.

The recommended provision of the bearer services is marked by essential (E1, E2, E3, Ex) or additional (A, Ax) as defined in rec. 01.06, section 2.2. The provision of the bearer services may differ from that of the bearer service category they belong to. This reflects, primarily, the necessity of providing specific interworking functions within the network for that bearer service, e.g V.32 modem for connection to PSTN.

The recommended provision of a bearer service in Table 3 when related to data services of 4.8kbit/s and below, with regard to the use of either a full or half rate channel is for further study.

Bearer services are numbered for ease of identification, the first two digits represent the basic attributes, the third digit represents the terminating network e.g PLMN, with the fourth digit representing the access at the mobile station and by inference, where necessary the access to the terminating network through the interworking function (see table 4). These numbers are not intended to be used for any other purpose.

Because the information transfer rate depends on the reference point assumed in the GSM PLMN, transit or terminating network, the attribute value is marked as "Not applicable".

Services not mentioned are not provided.

4. DESCRIPTION OF INDIVIDUAL BEARER SERVICES

Each bearer service category has been provided with a data sheet which contains values for all its attributes and comments and if appropriate annexes to it are contained in Appendix A.

5. LIST OF BEARER SERVICE CATEGORIES

NUMBER	DATA SHEET No	BEARER SERVICE CATEGORY NAME
1	1 With T	Circuit Mode Unstructured with Unrestricted digital capability Transparent
2	1 With NT	Circuit Mode Unstructured with Unrestricted digital capability Non Transparent
3	2	Pad services
4	3	Packet Service
5	4 With T	Circuit mode Unstructured with Alternate Speech /Unrestricted Digital capability Transparent
6	4 With NT	Circuit mode Unstructured with Alternate Speech /Unrestricted Digital capability Non Transparent
7	5 With T	Circuit mode Unstructured with Speech followed by unrestricted digital capability Transparent
8	5 With NT	Circuit mode Unstructured with Speech followed by unrestricted digital capability Non Transparent
9	6 With T	3.1 kHz Ex PLMN Transparent
10	6 With NT	3.1 kHz Ex PLMN Non Transparent

Table 2/GSM 02.02. List Of PLMN bearer service categories

6. LIST OF BEARER SERVICES

				Recommended provision in terminating network					
No	Bearer Service cat No	Bearer service Name	Quality of service attrib. value	PLMN	PSTN	ISDN	CSPDN	PSPDN	Direct Access
				0	1	2	3	4	5
21	1 9	Data circuit Duplex Asynchronus 300 bit/s	T	E2	E2	Ex			A
	2 10	Data circuit Duplex Asynchronus 300 bit/s	NT	E2	E2	Ex			A
22	1 9	Data circuit Duplex Asynchronus 1200 bit/s	T	E2	E2	Ex			A
	2 10	Data circuit Duplex Asynchronus 1200 bit/s	NT	E2	E2	Ex			A
23	1 9	Data circuit Duplex Asynchronus 1200/75 bit/s	T	FS	FS	FS			A
	2 10	Data circuit Duplex Asynchronus 1200/75 bit/s	NT	FS	FS	FS			A
24	1 9	Data circuit Duplex Asynchronus 2400 bit/s	T	A	A	Ex			A
	2 10	Data circuit Duplex Asynchronus 2400 bit/s	NT	A	A	Ex			A
25	1 9	Data circuit Duplex Asynchronus 4800 bit/s	T	A	A	Ex			A
	2 10	Data circuit Duplex Asynchronus 4800 bit/s	NT	A	A	Ex			A

26	1 9	Data circuit Duplex Asynchronus 9600 bit/s	T	A	A	Ex			A
	2 10	Data circuit Duplex Asynchronus 9600 bit/s	NT	A	A	Ex			A
31	1 9	Data circuit Duplex Synchronus 1200 bit/s	T	A	A	Ax			A
32	1 9	Data circuit Duplex Synchronus 2400 bit/s	T	A	A	Ax			A
33	1 9	Data circuit Duplex Synchronus 4800 bit/s	T	A	A	Ax			A
34	1 9	Data circuit Duplex Synchronus 9600 bit/s	T	A	A	Ax			A
41	3 9	PAD Access circuit Asynchronus 300 bit/s	T			FS		E2	A
	3 10	PAD Access circuit Asynchronus 300 bit/s	NT			FS		E2	A
42	3 9	PAD Access circuit Asynchronus 1200 bit/s	T			FS		E2	A
	3 10	PAD Access circuit Asynchronus 1200 bit/s	NT			FS		E2	A
43	3 9	PAD Access circuit Asynchronus 1200/75 bit/s	T			FS		A	A
	3 10	PAD Access circuit Asynchronus 1200/75 bit/s	NT			FS		A	A
44	3 9	PAD Access circuit Asynchronus 2400 bit/s	T			FS		A	A
	3 10	PAD Access circuit Asynchronus 2400 bit/s	NT			FS		A	A
45	3 9	PAD Access circuit Asynchronus 4800 bit/s	T			FS		A	A
	3 10	PAD Access circuit Asynchronus 4800 bit/s	NT			FS		A	A
46	3 9	PAD Access circuit Asynchronus 9600 bit/s	T			FS		A	A
	3 10	PAD Access circuit Asynchronus 9600 bit/s	NT			FS		A	A
51	4	Data Packet Duplex Synchronus 2400 bit/s	T	E3		FS		E3	FS
	4	Data Packet Duplex Synchronus 2400 bit/s	NT	FS		FS		FS	FS
52	4	Data Packet Duplex Synchronus 4800 bit/s	T	E3		FS		E3	FS
	4	Data Packet Duplex Synchronus 4800 bit/s	NT	FS		FS		FS	FS
53	4	Data Packet Duplex Synchronus 9600 bit/s	T	A		FS		A	FS
	4	Data Packet Duplex Synchronus 9600 bit/s	NT	FS		FS		FS	FS
61	5	Alternate Speech/Unrestrict Digital (unres. dig. section of service offers the same service as Services 21-34)	T for Unres dig	A	A	Ax			A

	6	Alternate Speech/Unrestrict Digital (unres. dig. section of service offers the same service as Services 21-34)	NT for Unres dig	A	A	Ax			A
71	1	12 kbit/s Unrestricted dig.		A					A
81	7	Speech followed by data (unres. dig. section of service offers the same service as Services 21-34)	T for Unres dig	A	A	A			A
	8	Speech followed by data (unres. dig. section of service offers the same service as Services 21-34)	NT for Unres dig	A	A	A			A

Table 3/GSM 02.02

Bearer services utilising a set of attributes within a bearer service category and their recommended provision

Access Interface at the MS attribute 9.2 and where necessary at the IWF	Code for fourth figure in bearer service No
Analogue 4 wire	0
V21 DTE/DCE interface	1
V22 DTE DCE interface	2
V22 bis DTE DCE interface	3
V23 DTE DCE interface	4
V26 ter DTE DCE interface	5
V32 DTE DCE interface	6
X21	7
X25	8
S interface	9
Undefined interface	a

Table 4/GSM 02.02

Coding of fourth figure in bearer service

Number	Recommended provision
2101	E2
2111	E2
2121	Ex
2151	A
2202	E2
2212	E2
2222	Ex
2252	A
2304	FS
2314	A
2324	FS
2354	FS
240A	A
2416	A
242A	FS
245A	A
250A	A
2516	A
252A	FS
255A	A
260A	A
2616	A
262A	FS
265A	A
3102	A
3112	A
3203	A
3205	A
3207	A
3213	A
3215	A
3227	Ax
3237	A
3306	A
3307	A
3316	A
3327	Ax
3337	A
3406	A
3407	A
3416	A
3427	Ax
3437	A
4121	FS
4141	E2
4151	A

4222	FS
4242	E2
4252	A
4324	FS
4344	A
4354	A
442A	FS
444A	A
445A	A
452A	FS
454A	A
455A	A
462A	FS
464A	A
465A	A
5108	E3
5128	FS
5148	E3
5158	FS
5208	E3
5228	FS
5248	E3
5258	FS
5308	A
5328	FS
5348	A
5358	FS
6101	A
6102	A
6103	A
6104	A
6105	A
6106	A
6109	A
611(1-9 AS ABOVE)	A
612(1-9 AS ABOVE)	Ax
615(1-9 AS ABOVE)	A
710A	A
715A	A
81** (same as service 61)	A

Table 5/GSM 02.02

List of GSM bearer services

Annex A

(to recommendation GSM 02.02)

Description of individual bearer services

BEARER SERVICE CATEGORY 1

CIRCUIT MODE UNSTRUCTURED WITH UNRESTRICTED DIGITAL CAPABILITY TRANSPARENT

1) DEFINITION

An unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) Sub-rate information streams which are rate adapted.

ii) Transparent support of services i.e Quality of service attribute is set for transparent value.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

BEARER SERVICE CATEGORY 2

CIRCUIT MODE UNSTRUCTURED WITH UNRESTRICTED DIGITAL CAPABILITY NON TRANSPARENT

1) DEFINITION

An unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) Sub-rate information streams which are rate adapted.

ii) Non transparent support of services i.e Quality of service attribute is set for non transparent value.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

2) ATTRIBUTE VALUES

Data sheet No 1	TITLE OF BEARER SERVICE : **Circuit mode Unstructured with unrestricted digital capability transparent /non transparent**
1	INFORMATION TRANSFER CIRCUIT [X] PACKET [ ]
2	INFORMATION TRANSFER RATE Not Applicable
3	INFORMATION TRANSFER CAPABILITY Unrestricted digital [X] speech [ ] 3.1kHz Ex PLMN [X] Alternate speech /group 3 facsimile [ ] Alternate speech /Unres Digital [ ] Speech followed by Unres Digital [ ]
4	STRUCTURE Structured [ ] Service data unit integrity [ ] Unstructured [X]
5	ESTABLISHMENT OF COMMUNICATION Demand [X]
6	COMMUNICATION CONFIGURATION Point to point [X] Point to multipoint [ ]
7	SYMMETRY Unidirectional [ ] Bi-Directional symmetric [X] Bi-Directional asymmetric [ ]

	TITLE OF BEARER SERVICE : Circuit mode Unstructured with unrestricted digital capability																				NUMBER 1
8	ACCESS CHANNEL AND RATE (at access point 1/2)																				
	12k bit/s	9.6kbit/s				4.8kbit/s			2.4kbit/s				1.2kbit/s			1200/75 bit/s		300bit/s		64kbit/s (MIC	
9.1	SIGNALLING ACCESS (at access points 1/2)																				
manual	X	X	X					X	X			X	X	X	X		X		X		
X21				X			X				X										
I440/450																					X
V25 bis																X					
X32																					
X28																					
9.2	INFORMATION ACCESS PROTOCOL (at access point 1/2)																				
G711																					
I450/451																					X
X25																					
V22 bis													X								
V26 Ter												X									
V32		X						X													
X21				X			X				X										
V21																				X	
V22			X[1]						X[1]					X[1]	X	X					
V23																	X				
X28																					
X29																					
9.3	INFORMATION ACCESS STRUCTURE (at access point 1/2)																				
SYNCHR		X		X			X	X		X	X	X	X		X	X					X
ASYNCH			X						X					X	X	X	X		X		X

[1]
Note 1 : The V22 DTE/DCE interface is used as an example of this type of access. Any DTE/DCE interface could be used which has an Asynchronous capability, with the user rate changed where necessary.

Data sheet No 1	TITLE OF BEARER SERVICE : **Circuit mode Unstructured with unrestricted digital capability transparent /non transparent**
10	SUPPLEMENTARY SERVICES The description of these services are contained in recommendation 02.04
11	QUALITY OF SERVICE This attribute has two sets of values, one for the transparent support of data services the other for the non-transparent support of data services. For the definition of transparent and non-transparent concepts refer to recommendation 01.04 and 03.10. For the quality of service parameters associated with both options refer to recommendation 02.08. Transparent / Non Transparent [X] Asynchronous [X] Transparent / Non Transparent [X] Synchronous [ ]
12	INTERWORKING POSSIBILITIES Intercommunication with services in the PSTN, PSPDN, ISDN, CSPDN, PLMN and DIRECT ACCESS is required, for specific applications refer to Table 3. Changes to the capabilities as a result of this may be necessary. Interworking capabilities are contained in recommendations 09.01, 09.02, 09.03, 09.04, 09.05, 09.06 and 09.07.
13	OPERATIONAL COMMERCIAL

BEARER SERVICE CATEGORY 3

PAD SERVICES

1) DEFINITION

An unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) Sub-rate information streams which are rate adapted.

ii) Access to a packet assembler/disassembler function.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

2) ATTRIBUTE VALUES

Data sheet No 1	TITLE OF BEARER SERVICE : **Pad services**
1	INFORMATION TRANSFER CIRCUIT [X] PACKET [ ]
2	INFORMATION TRANSFER RATE Not Applicable
3	INFORMATION TRANSFER CAPABILITY Unrestricted digital [X] speech [ ] 3.1kHz Ex PLMN [X] Alternate speech /group 3 facsimile [ ] Alternate speech /Unres Digital [ ] Speech followed by Unres Digital [ ]
4	STRUCTURE Structured [ ] Service data unit integrity [X] Unstructured [ ]
5	ESTABLISHMENT OF COMMUNICATION Demand [X]
6	COMMUNICATION CONFIGURATION Point to point [X] Point to multipoint [ ]
7	SYMMETRY Unidirectional [ ] Bi-Directional symmetric [x] Bi-Directional asymmetric [x]

	TITLE OF BEARER SERVICE : Pad services																			NUMBER 2	
	ACCESS CHANNEL AND RATE (at access point 1/2)																				
	12k bit/s		9.6kbit/s		4.8kbit/s		2.4kbit/s		1.2kbit/s		1200/75 bit/s		300bit/s						64kbit/s (via MIC)		
9.1	SIGNALLING ACCESS (at access points 1/2)																				
manual		X		X		X		X		X		X									
X21																					
I440/450																			X		
V25 bis																					
X32																					
X28		X		X		X		X		X		X									
9.2	INFORMATION ACCESS PROTOCOL (at access point 1/2)																				
G711																					
I450/451																			X		
X25																					
V22 bis																					
V26 Ter																					
V32																					
X21																					
V21													X								
V22		X[2]		X[1]		X[1]		X													
V23										X											
X28																					
X29																					
9.3	INFORMATION ACCESS STRUCTURE (at access point 1/2)																				
SYNCHR																					
ASYNCH		X		X		X		X		X		X							X		

2
Note 1 : The V22 DTE/DCE interface is used as an example of this type of access. Any DTE/DCE interface could be used which has an Asynchronous capability, with the user rate changed where necessary.

Data sheet No 2	TITLE OF BEARER SERVICE : **Pad services**
10	SUPPLEMENTARY SERVICES The description of these services are contained in recommendation 02.04
11	QUALITY OF SERVICE This attribute has two sets of values. one for the transparent support of data services the other for the non-transparent support of data services. For the definition of transparent and non-transparent concepts refer to recommendation 01.04 and 03.10. For the quality of service parameters associated with both options refer to recommendation 02.08. Transparent: [X] Non Transparent: [X]
12	INTERWORKING POSSIBILITIES Intercommunication with services in the PSTN, PSPDN, ISDN, CSPDN, PLMN and DIRECT ACCESS is required, for specific applications refer to Table 3. Changes to the capabilities as a result of this may be necessary. Interworking capabilities are contained in recommendations 09.01, 09.02, 09.03, 09.04, 09.05, 09.06 and 09.07.
13	OPERATIONAL COMMERCIAL

BEARER SERVICE CATEGORY 4

PACKET SERVICE

1) DEFINITION

An unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) Access to a X25 public data network (Recommendation 07.03,) using X32 or X31 case A access

ii) Utilisation of X31 case B access (Virtual circuit bearer service)

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

2) ATTRIBUTE VALUES

Data sheet No 3	TITLE OF BEARER SERVICE : **Packet mode service**
1	INFORMATION TRANSFER (Note: circuit if using X32/X31 case A : packet if using X31 case B) CIRCUIT [X] PACKET [X]
2	INFORMATION TRANSFER RATE
3	INFORMATION TRANSFER CAPABILITY Unrestricted digital [X] speech [ ] 3.1kHz Ex PLMN [ ] Alternate speech /group 3 facsimile [ ] Alternate speech /Unres Digital [ ] Speech followed by Unres Digital [ ]
4	STRUCTURE Structured [ ] Service data unit integrity [X] Unstructured [ ]
5	ESTABLISHMENT OF COMMUNICATION Demand [X]
6	COMMUNICATION CONFIGURATION Point to point [X] Point to multipoint [ ]
7	SYMMETRY Unidirectional [ ] Bi-Directional symmetric [X] Bi-Directional asymmetric [ ]

	TITLE OF BEARER SERVICE : Packet service																	NUMBER 3		
8	ACCESS CHANNEL AND RATE (at access point 1/2)																			
	12 kbit/s		9.6kbit/s		4.8kbit/s		2.4kbit/s		1.2kbit/s		1200/75bit/s		300bit/s					64kbit/s (via MIC)		
9.1	SIGNALLING ACCESS (at access points 1/2)																			
manual			X		X		X													
X21																				
I440/450																			X	
V25 bis																				
X32				X		X		X												
X28																				
9.2	INFORMATION ACCESS PROTOCOL (at access point 1/2)																			
G711																				
I450/451																			X	
X25			X	X	X	X	X	X												
V22 bis																				
V26 Ter																				
V32																				
X21																				
V21																				
V22																				
V23																				
X28																				
X29																				
9.3	INFORMATION ACCESS STRUCTURE (at access point 1/2)																			
SYNCHR			X	X	X	X	X	X											X	
ASYNCH																				

Data sheet No 3	TITLE OF BEARER SERVICE : **Packet mode service**
10	SUPPLEMENTARY SERVICES The description of these services are contained in recommendation 02.04
11	QUALITY OF SERVICE This attribute has two sets of values, one for the transparent support of data services the other for the non-transparent support of data services. For the definition of transparent and non-transparent concepts refer to recommendation 01.04 and 03.10. For the quality of service parameters associated with both options refer to recommendation 02.08. Transparent: [X] Non Transparent: [X]
12	INTERWORKING POSSIBILITIES Intercommunication with services in the PSTN, PSPDN, ISDN, CSPDN, PLMN and DIRECT ACCESS is required, for specific applications refer to Table 3. Changes to the capabilities as a result of this may be necessary. Interworking capabilities are contained in recommendations 09.01, 09.02, 09.03, 09.04, 09.05, 09.06 and 09.07.
13	OPERATIONAL COMMERCIAL

BEARER SERVICE CATEGORY 5

CIRCUIT MODE UNSTRUCTURED WITH ALTERNATE SPEECH UNRESTRICTED DIGITAL CAPABILITY TRANSPARENT

1) DEFINITION

An Alternate speech/unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) The unrestricted digital capability allows the use of sub-rate information streams which are rate adapted.

ii) The capability to swap between speech and data during a call.

iii) Transparent support of services i.e the quality of service attribute value is set to transparent.

It is intended that where either the speech or data service requires the use of a full rate channel and the other service requires the use of a half rate channel, that a full rate channel will be reserved for the duration of the call, with a half f rate channel remaining unused for the period where the service only requires a half rate channel.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

BEARER SERVICE CATEGORY 6

CIRCUIT MODE UNSTRUCTURED WITH ALTERNATE SPEECH UNRESTRICTED DIGITAL CAPABILITY NON TRANSPARENT

1) DEFINITION

An Alternate speech/unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) The unrestricted digital capability allows the use of sub-rate information streams which are rate adapted.

ii) The capability to swap between speech and data during a call.

iii) Non transparent support of services i.e the quality of service attribute value is set to non transparent.

It is intended that where either the speech or data service requires the use of a full rate channel and the other service requires the use of a half rate channel, that a full rate channel will be reserved for the duration of the call, with a half f rate channel remaining unused for the period where the service only requires a half rate channel.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

2) ATTRIBUTE VALUES

Data sheet No 4	TITLE OF BEARER SERVICE : **Circuit mode Unstructured with alternate speech/unrestricted digital capability transparent/non transparent**
1	INFORMATION TRANSFER CIRCUIT [X] PACKET [ ]
2	INFORMATION TRANSFER RATE Not Applicable
3	INFORMATION TRANSFER CAPABILITY Unrestricted digital [ ] speech [ ] 3.1kHz Ex PLMN [ ] Alternate speech /group 3 facsimile [ ] Alternate speech /Unres Digital [X] Speech followed by Unres Digital [ ]
4	STRUCTURE Structured [ ] Service data unit integrity [ ] Unstructured [X]
5	ESTABLISHMENT OF COMMUNICATION Demand [X]
6	COMMUNICATION CONFIGURATION Point to point [X] Point to multipoint [ ]
7	SYMMETRY Unidirectional [ ] Bi-Directional symmetric [X] Bi-Directional asymmetric [ ]

	TITLE OF BEARER SERVICE : CCT MODE UNST WITH ALT SPEECH/UNRES DIG TRAN/NON TRAN																			NUMBER 4
8	ACCESS CHANNEL AND RATE (at access point 1/2)																			
	12k bit/s	9.6kbit/s		4.8kbit/s		2.4kbit/s				1.2kbit/s		1200/75 bit/s			300bit/s			64kbit/s (via MIC)		
9.1	SIGNALLING ACCESS (at access points 1/2)																			
manual		X	X	X	X		X	X	X	X		X			X					
X21						X													X	
I440/450											X									
V25 bis																				
X32																				
X28																				
9.2	INFORMATION ACCESS PROTOCOL (at access point 1/2)																			
G711																			X	
I450/451																				
X25																				
V22 bis								X												
V26 Ter					X		X													
V32		X																		
X21						X														
V21															X					
V22			X[1]	X[1]					X[3]	X	X									
V23												X								
X28																				
X29																				
9.3	INFORMATION ACCESS STRUCTURE (at access point 1/2)																			
SYNCHR		X			X	X	X	X		X	X								X	
ASYNCH			X	X					X	X	X	X			X				X	

3
Note 1 : The V22 DTE/DCE interface is used as an example of this type of access. Any DTE/DCE interface could be used which has an Asynchronous

Data sheet No 4	TITLE OF BEARER SERVICE : **Circuit mode Unstructured with alternate speech/unrestricted digital capability transparent/non transparent**
10	SUPPLEMENTARY SERVICES The description of these services are contained in recommendation 02.04
11	QUALITY OF SERVICE This attribute has two sets of values, one for the transparent support of data services the other for the non-transparent support of data services. For the definition of transparent and non-transparent concepts refer to recommendation 01.04 and 03.10. For the quality of service parameters associated with both options refer to recommendation 02.08. Transparent Non Transparent [X] Asynchronous [X] Transparent Non Transparent [X] Synchronous [ ]
12	INTERWORKING POSSIBILITIES Intercommunication with services in the PSTN, PSPDN, ISDN, CSPDN, PLMN and DIRECT ACCESS is required, for specific applications refer to Table 3. Changes to the capabilities as a result of this may be necessary. Interworking capabilities are contained in recommendations 09.01, 09.02, 09.03, 09.04, 09.05, 09.06 and 09.07.
13	OPERATIONAL COMMERCIAL

Note : This bearer service will utilise a full rate channel for the duration of the call (if either the speech capability or data capability indicates it as necessary), i.e either the speech or data portion of the call may only utilise a half rate channel, however the full channel capability should be retained.

: The access interface at the mobile station is assumed to be different for the data and the speech usage. Some means must be provided to select the speech/data capability.

BEARER SERVICE CATEGORY 7

CIRCUIT MODE UNSTRUCTURED WITH SPEECH FOLLOWED BY UNRESTRICTED DIGITAL CAPABILITY TRANSPARENT

1) DEFINITION

A speech followed by unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) The unrestricted digital capability allows the use of Sub-rate information streams which have been rate adapted.

ii) With this service the intention is to establish a speech call in the first instance and then at some time while the call is in progress for the user to be able to switch to the data capability. It is intended that if this swap causes a change from a full channel usage to a half rate channel usage, that the remaining half rate channel will be released i.e the user is unable to swap back again.

iii) Transparent support of services i.e quality of service attribute value is set for transparent.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

BEARER SERVICE CATEGORY 8

CIRCUIT MODE UNSTRUCTURED WITH SPEECH FOLLOWED BY UNRESTRICTED DIGITAL CAPABILITY NON TRANSPARENT

1) DEFINITION

A speech followed by unrestricted bearer service provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) The unrestricted digital capability allows the use of Sub-rate information streams which have been rate adapted.

ii) With this service the intention is to establish a speech call in the first instance and then at some time while the call is in progress for the user to be able to switch to the data capability. It is intended that if this swap causes a change from a full channel usage to a half rate channel usage, that the remaining half rate channel will be released i.e the user is unable to swap back again.

iii) Non transparent support of services i.e quality of service attribute value is set for non transparent.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

2) ATTRIBUTE VALUES

Data sheet No 5	TITLE OF BEARER SERVICE : **Circuit mode Unstructured with speech followed by unrestricted digital capability transparent/non transparent**
1	INFORMATION TRANSFER CIRCUIT [X] PACKET [ ]
2	INFORMATION TRANSFER RATE Not Applicable
3	INFORMATION TRANSFER CAPABILITY Unrestricted digital [ ] speech [ ] 3.1kHz Ex PLMN [ ] Alternate speech /group 3 facsimile [ ] Alternate speech /Unres Digital [ ] Speech followed by Unres Digital [X]
4	STRUCTURE Structured [ ] Service data unit integrity [ ] Unstructured [X]
5	ESTABLISHMENT OF COMMUNICATION Demand [X]
6	COMMUNICATION CONFIGURATION Point to point [X] Point to multipoint [ ]
7	SYMMETRY Unidirectional [ ] Bi-Directional symmetric [X] Bi-Directional asymmetric [ ]

	TITLE OF BEARER SERVICE : CCT MODE UNSTR WITH SPEECH FOLLOWED BY UNRES DIG CAPABILITY TRANS/NON TRANS																			NUMBER 5
8	ACCESS CHANNEL AND RATE (at access point 1/2)																			
	12k bit/s	9.6kbit/s		4.8kbit/s		2.4kbit/s				1.2kbit/s		1200/75 bit/s			300bit/s			64kbit/s (via MIC)		
9.1	SIGNALLING ACCESS (at access points 1/2)																			
manual		X	X	X	X		X	X	X	X		X			X					
X21						X														
I440/450																			X	
V25 bis											X									
X32																				
X28																				
9.2	INFORMATION ACCESS PROTOCOL (at access point 1/2)																			
G711																				
I450/451																			X	
X25																				
V22 bis								X												
V26 Ter					X		X													
V32		X																		
X21						X														
V21															X					
V22			X[1]	X[1]					X[4]	X	X									
V23												X								
X28																				
X29																				
9.3	INFORMATION ACCESS STRUCTURE (at access point 1/2)																			
SYNCHR		X			X	X	X	X		X	X								X	
ASYNCH			X	X					X	X	X	X			X				X	

4
Note 1 : The V22 DTE/DCE interface is used as an example of this type of access. Any DTE/DCE interface could be used which has an Asynchronous capability, with the user rate changed where necessary.

Data sheet No 5	TITLE OF BEARER SERVICE : **Circuit mode Unstructured with speech followed by unrestricted digital capability transparent/non transparent**
10	SUPPLEMENTARY SERVICES The description of these services are contained in recommendation 02.04
11	QUALITY OF SERVICE This attribute has two sets of values, one for the transparent support of data services the other for the non-transparent support of data services. For the definition of transparent and non-transparent concepts refer to recommendation 01.04 and 03.10. For the quality of service parameters associated with both options refer to recommendation 02.08. Transparent — Non Transparent [X] Asynchronous [X] Transparent — Non Transparent [X] Synchronous [ ]
12	INTERWORKING POSSIBILITIES Intercommunication with services in the PSTN, PSPDN, ISDN, CSPDN, PLMN and DIRECT ACCESS is required, for specific applications refer to Table 3. Changes to the capabilities as a result of this may be necessary. Interworking capabilities are contained in recommendations 09.01, 09.02, 09.03, 09.04, 09.05, 09.06 and 09.07.
13	OPERATIONAL COMMERCIAL

BEARER SERVICE CATEGORY 9

3.1 KHZ EX PLMN TRANSPARENT

1) DEFINITION

A Bearer Service based on the 3.1 kHz Ex PLMN bearer service category provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) This service supports the use of Sub-rate information streams which have been rate adapted.

ii) The capability to the requirement to select a modem at the interworking function.

iii) Transparent support of services i.e quality of service attribute value is set for transparent.

It is intended that this service should utilise an unrestricted digital connection within the PLMN, but that it should be used to select the "3.1 kHz Audio" service when interworking to the ISDN. It is not intended that this service should support the transmission of speech or voiceband data directly, within the PLMN, as the ISDN bearer service does within the ISDN.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

BEARER SERVICE CATEGORY 10

3.1 KHZ EX PLMN NON TRANSPARENT

1) DEFINITION

A Bearer Service based on the 3.1 kHz Ex PLMN bearer service category provides information transfer between R/S reference points. It may therefore be used to support various user applications. Examples include:

i) This service supports the use of Sub-rate information streams which have been rate adapted.

ii) The capability to the requirement to select a modem at the interworking function.

iii) Non transparent support of services i.e quality of service attribute value is set for non transparent.

It is intended that this service should utilise an unrestricted digital connection within the PLMN, but that it should be used to select the "3.1 kHz Audio" service when interworking to the ISDN. It is not intended that this service should support the transmission of speech or voiceband data directly, within the PLMN, as the ISDN bearer service does within the ISDN.

User information is transferred over a Bm or Lm channel; signalling is provided over a Dm channel.

2) ATTRIBUTE VALUES

Data sheet No 6	TITLE OF BEARER SERVICE : **3.1 kHz Ex PLMN transparent /non transparent**
1	INFORMATION TRANSFER CIRCUIT [X] PACKET [ ]
2	INFORMATION TRANSFER RATE
3	INFORMATION TRANSFER CAPABILITY Unrestricted digital [ ] speech [ ] 3.1kHz Ex PLMN [X] Alternate speech /group 3 facsimile [ ] Alternate speech /Unres Digital [ ] Speech followed by Unres Digital [ ]
4	STRUCTURE Structured [ ] Service data unit integrity [ ] Unstructured [X]
5	ESTABLISHMENT OF COMMUNICATION Demand [X]
6	COMMUNICATION CONFIGURATION Point to point [X] Point to multipoint [ ]
7	SYMMETRY Unidirectional [ ] Bi-Directional symmetric [X] Bi-Directional asymmetric [ ]

	TITLE OF BEARER SERVICE : 3.1 KHZ EX PLMN TRANS/NON TRANS																			NUMBER 6
8	ACCESS CHANNEL AND RATE (at access point 1/2)																			
	12k bit/s	9.6kbit/s		4.8kbit/s		2.4kbit/s				1.2kbit/s		1200/75 bit/s			300bit/s			64kbit/s (via MIC)		
9.1	SIGNALLING ACCESS (at access points 1/2)																			
manual		X	X	X	X		X	X	X	X		X			X					
X21						X													X	
I440/450											X									
V25 bis																				
X32																				
X28																				
9.2	INFORMATION ACCESS PROTOCOL (at access point 1/2)																			
G711																			X	
I450/451																				
X25																				
V22 bis								X												
V26 Ter					X		X													
V32		X																		
X21						X														
V21															X					
V22			X[1]	X[1]					X[5]	X	X									
V23												X								
X28																				
X29																				
9.3	INFORMATION ACCESS STRUCTURE (at access point 1/2)																			
SYNCHR		X			X	X	X	X		X	X								X	
ASYNCH			X	X					X	X	X	X			X				X	

5
Note 1 : The V22 DTE/DCE interface is used as an example of this type of access. Any DTE/DCE interface could be used which has an Asynchronous

Data sheet No 6	TITLE OF BEARER SERVICE : **3.1 kHz Ex PLMN transparent /non transparent**
10	SUPPLEMENTARY SERVICES The description of these services are contained in recommendation 02.04
11	QUALITY OF SERVICE This attribute has two sets of values, one for the transparent support of data services the other for the non-transparent support of data services. For the definition of transparent and non-transparent concepts refer to recommendation 01.04 and 03.10. For the quality of service parameters associated with both options refer to recommendation 02.08. Transparent Non Transparent [X] Asynchronous [X] Transparent Non Transparent [X] Synchronous [ ]
12	INTERWORKING POSSIBILITIES Intercommunication with services in the PSTN, PSPDN, ISDN, CSPDN, PLMN and DIRECT ACCESS is required, for specific applications refer to Table 3. Changes to the capabilities as a result of this may be necessary. Interworking capabilities are contained in recommendations 09.01, 09.02, 09.03, 09.04, 09.05, 09.06 and 09.07.
13	OPERATIONAL COMMERCIAL